Sahaj Bhadja

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Summary

Software Engineer with 2+ years delivering **system-level testing**, **software/hardware validation workflows**, and **cross-functional product documentation** using Python, C++/Java, SQL, and cloud-connected development environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Executed **hands-on performance testing** and system validation for 500+ users, reducing latency from **1.2s to 350ms** through iterative tuning of software, data flows, and component behavior.
- Built **maturation-style validation workflows** for 5K+ monthly jobs, documenting outcomes and failures to improve product acceptance, traceability, and communication with technical stakeholders.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed and validated **Flask + React features** at 10K+ daily requests, translating feature requirements into testable component behavior and reliable user-facing functionality.
- Improved **release quality** via CI/CD controls and concurrent processing checks, reducing turnaround time by **60%** and strengthening confidence in production behavior under load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **Python simulation pipelines** over 100K+ trajectory logs, enabling repeatable component/configuration testing, performance analysis, and early prediction of behavior degradations.
- Produced clear technical reports and debugging summaries, improving engineering decision speed during design discussions and downstream implementation planning.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API/ETL components** using OOP, AWS Glue, and S3, improving response times by **30%** while stabilizing cross-system communication and configuration correctness.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** across three OS environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **REST API and WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving reliability of integrated software components in real operational environments.
- Automated deployment/testing using **Docker** and Jenkins, reducing deployment time by **50%** and strengthening release readiness through reproducible execution and validation procedures.

Skills

Languages: **Python**, **C++**, Java, SQL, JavaScript/TypeScript
System Development: requirements capture, component specification support, hardware/software interface validation mindset
Testing & Analysis: performance testing, maturation/configuration testing, root-cause analysis, report generation, failure prediction
Platform & Tools: Linux/UNIX basics, Git, CI/CD, Docker, Jenkins, cloud-connected environments
Engineering Practices: Agile/Scrum, code reviews, technical documentation, cross-functional communication, continuous improvement

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **six scheduling algorithms** and interactive diagnostics across 1K+ process combinations, enabling comparative performance analysis and validation of system behavior under varied runtime conditions.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed **API-driven operational modules** supporting 100+ daily appointments, adding monitoring checks that improved record retrieval by **2x** and strengthened reliability in user-facing workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built **integrated workflow services** for 50K+ users and 1M+ transactions with secure APIs, real-time updates, and scalable deployment behavior suitable for configuration-heavy systems.